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Course Code : EE 4121

VLSI

Introduction to VLSI Technology

Choice of VLSI Technology

- BJT, MOS (pMOS, nMOS), CMOS, BiCMOS

❖BJT

- Base current limits integration density
- More static power dissipation

❖MOS – low power allows very high integration

❑ Why MOS not BJT in integration?

Features	BJT	MOS
Speed	High	Low
Area	More	Less
Power Dissipation	More	Less
No. of Seps(MASKs)	More	Less
Processing	Complex	Simple
Repeatability	No	Yes
Input impedance	Low	High
Noise Margin	low	High
Output Driven Current	High	Low

* Positive Temperature co-efficient- MOS

❑NMOS or CMOS?

NMOS	CMOS
One transistor	Two transistor
Less area	More area
Low speed	High speed
High power dissipation	Low power dissipation
Complexity few	Less complexity
Less reliable	More reliable

❑Speed-Power Product:

❖BJT – Speed = 10ns, Power = 2mW, Speed*Power = 20 pJ

❖MOS – Speed = 40ns, Power = 10nW, Speed*Power = 0.0004 pJ

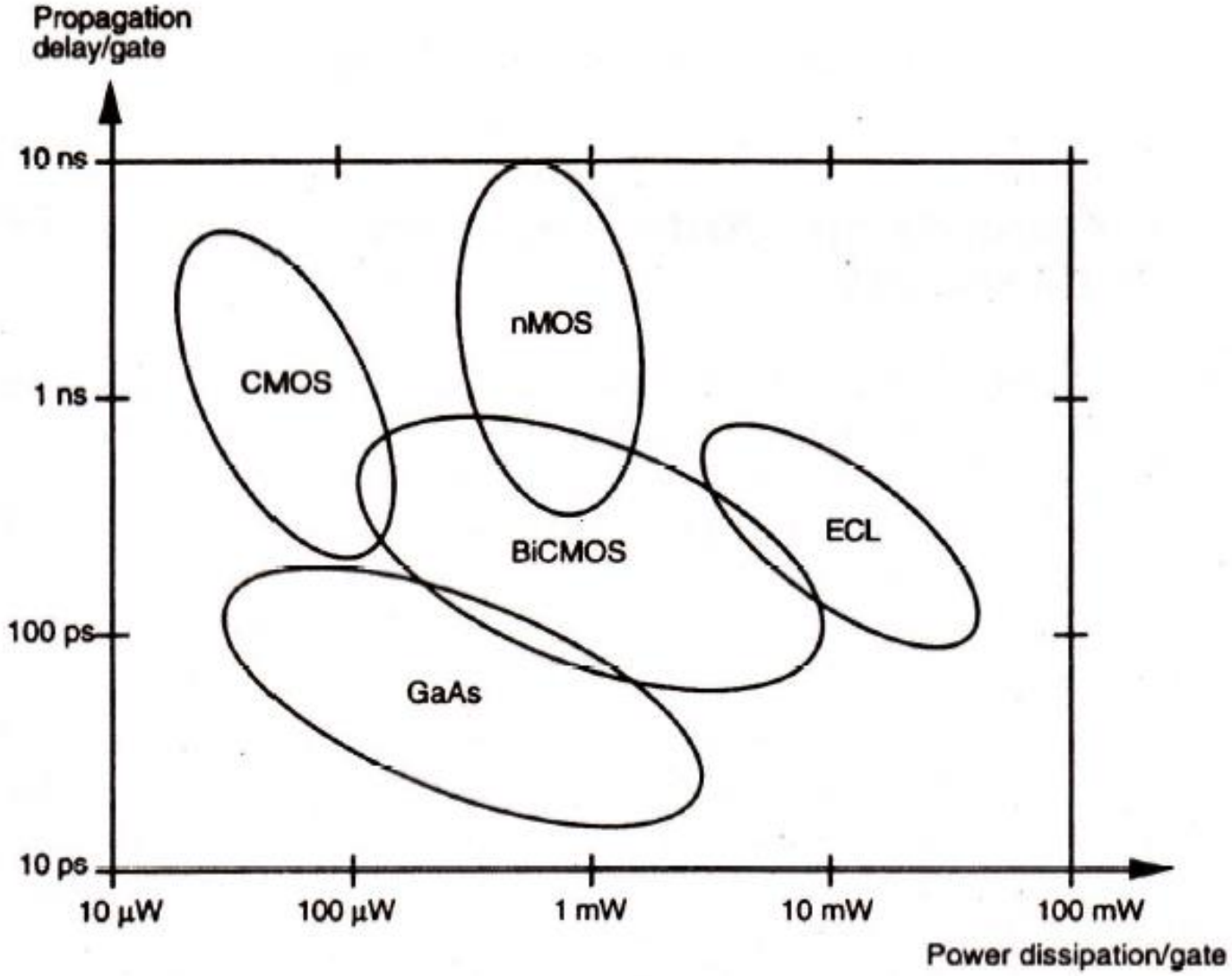


FIGURE 1.2 Speed/power performance of available technologies.

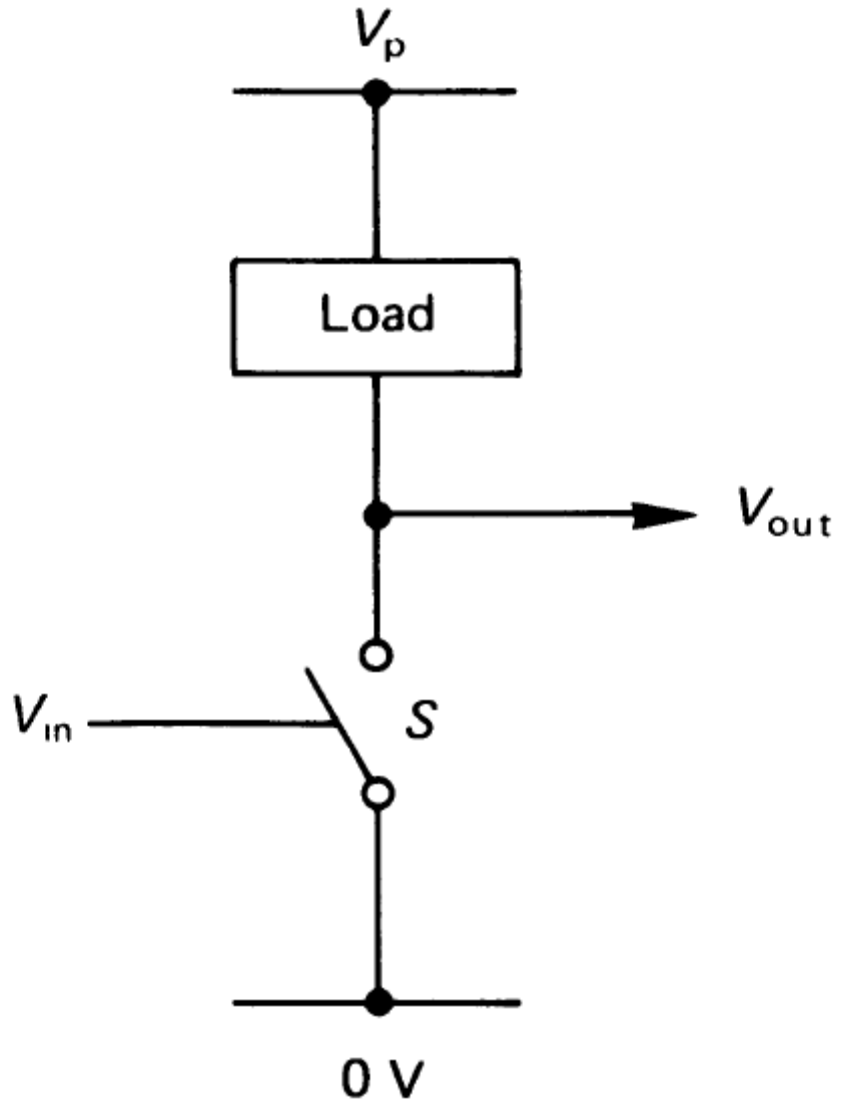


Figure 2.10 Inverter principle

2.7 NMOS Inverter with a Resistor Load

$$I_{sd} = \frac{\epsilon \mu_p W}{DL}\left[(V_{sg} - V_t)V_{sd} - \frac{V_{sd}^2}{2}\right]$$

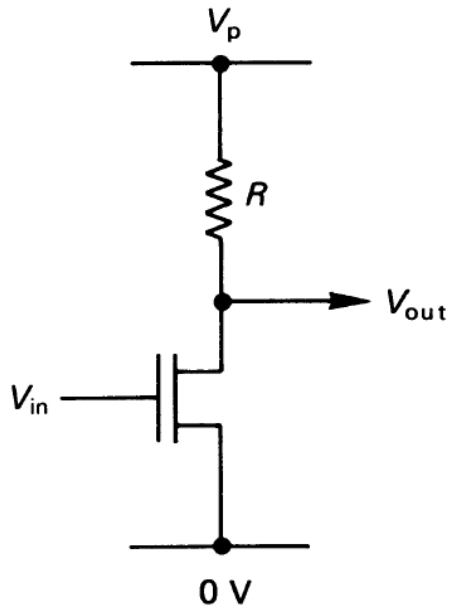


Figure 2.11 NMOS inverter with a resistor load

R is calculated to give a low V_{out} of 0.3 V_{te}, since this gives a reasonable noise margin between V_{out} and the gate threshold. Thus V_{ds} of the transistor is 0.3 V and, since V_{gs} is 5 V, $V_{ds} < V_{gs} - V_{te}$ and the device is operating in the resistive region. Equation (2.2) can now be used to find that

$$I_{ds} = 30\left(\frac{1}{1}\right)\left[(5-1)\,0.3 - \frac{0.3^2}{2}\right] = 34.7\ \mu A$$

This current also flows through the resistor. Hence

$$R = \frac{V_p - V_{out}}{I_{ds}} = \frac{5 - 0.3\ V}{34.7\ \mu A} = 135.4\ k\Omega$$

The silicon area required to implement this resistor is far larger than that for the transistor (typically by a factor of 300). Thus it is not practical to use a resistor as a load and so an MOS device is used instead.

2.8 NMOS Inverter with an NMOS Enhancement Transistor Load

The main drawback of an NMOS enhancement load, apart from the loss of voltage for high outputs, is the speed of the rising edge of the gate. This can be appreciated by considering that there is some capacitance C_{out} on V_{out}, and V_{out} is at 0.3 V corresponding to a V_{in} of 3.12 V. If V_{in} now switches to 0.3 V, T1 turns off and V_{out} starts to rise. Initially, V_{gs} of T2 is 4.7 V but as V_{out} rises, V_{gs} of T2 decreases and T2's threshold rises (because of the body effect) so that less current flows in T2. This progressively slows down the rate of charging up C_{out}, and as V_{out} approaches 3.12 V, T2 tends to turn off and very little current flows to finish the charging of C_{out}. For this reason, n-channel depletion devices are preferred as a load.

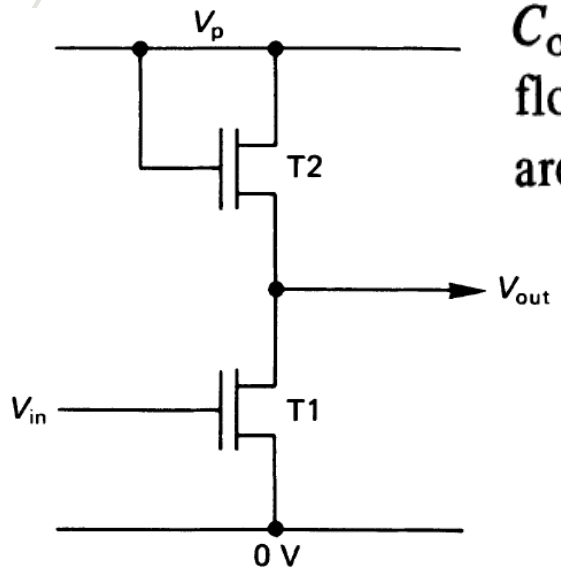


Figure 2.12 NMOS inverter with NMOS enhancement load

2.9 NMOS Inverter with an NMOS Depletion Transistor Load

Referring to figure 2.13, the NMOS depletion transistor T2 forms the circuit load. The depletion threshold V_{td} of T2 is negative and since the gate-source voltage of T2 is zero, T2 is always on.

When V_{in} is low, it is less than the enhancement threshold V_{te} of T1, and T1 is off. T2 is on and supplies the very small leakage current of T1. Since I_{ds} of T2 is negligible, the drain-source voltage of T2 is very small, leading to a V_{out} of V_p. Thus there is no high level output voltage loss as with the NMOS enhancement load.

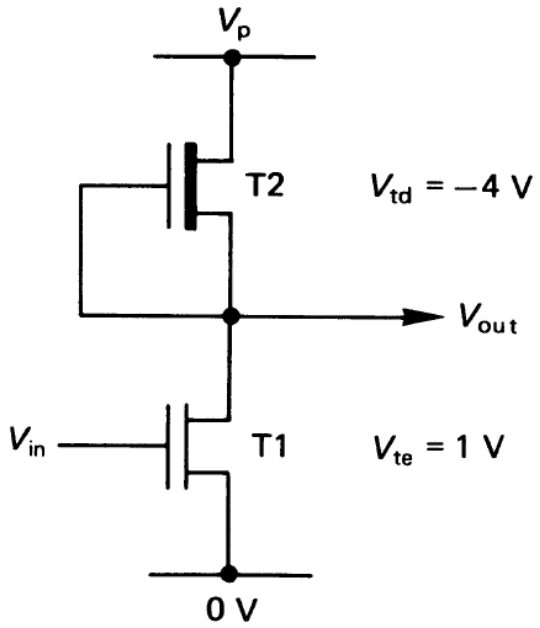


Figure 2.13 NMOS inverter with NMOS depletion load

2.12 The CMOS Inverter

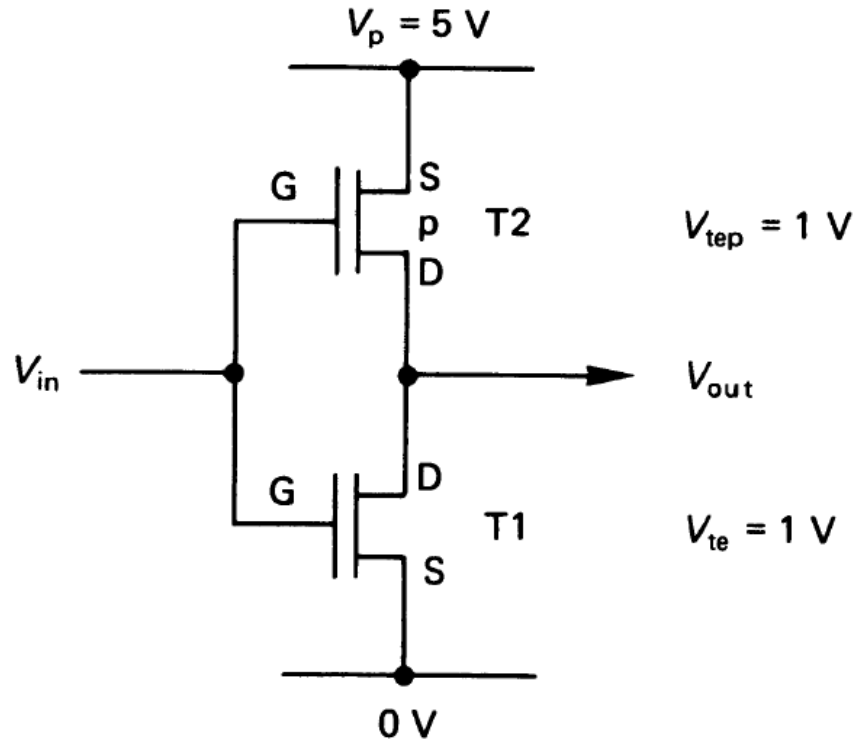


Figure 2.19 CMOS inverter

T1 and T2 both act as drivers and accept or supply current to charge the load capacitance when the gate switches. Edge times in CMOS approximate to the falling edge time for an NMOS inverter with a depletion load. Thus